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**Sir et al.**

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(54) **STRIPPED REDISTRUBUTION-LAYER  
FABRICATION FOR PACKAGE-TOP  
EMBEDDED MULTI-DIE INTERCONNECT  
BRIDGE**

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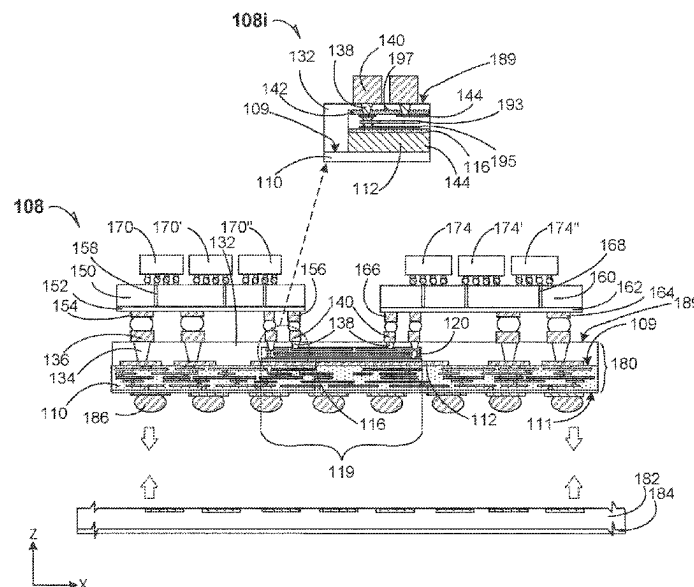
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(57) **ABSTRACT**

An embedded multi-die interconnect bridge (EMIB) is fab-  
ricated on a substrate using photolithographic techniques,  
and the EMIB is separated from the substrate and placed on  
the penultimate layer of an integrated-circuit package sub-  
strate, below the top solder-resist layer. A low Z-height of  
the EMIB, allows for useful trace and via real estate below  
the EMIB, to be employed in the package substrate.

**20 Claims, 6 Drawing Sheets**



**Related U.S. Application Data**

continuation of application No. 17/200,700, filed on Mar. 12, 2021, now Pat. No. 11,658,111, which is a continuation of application No. 16/384,348, filed on Apr. 15, 2019, now Pat. No. 10,998,262.

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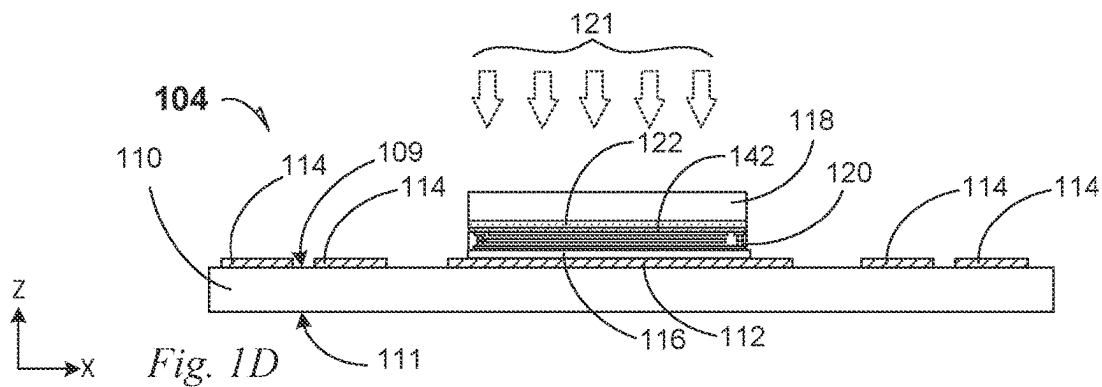
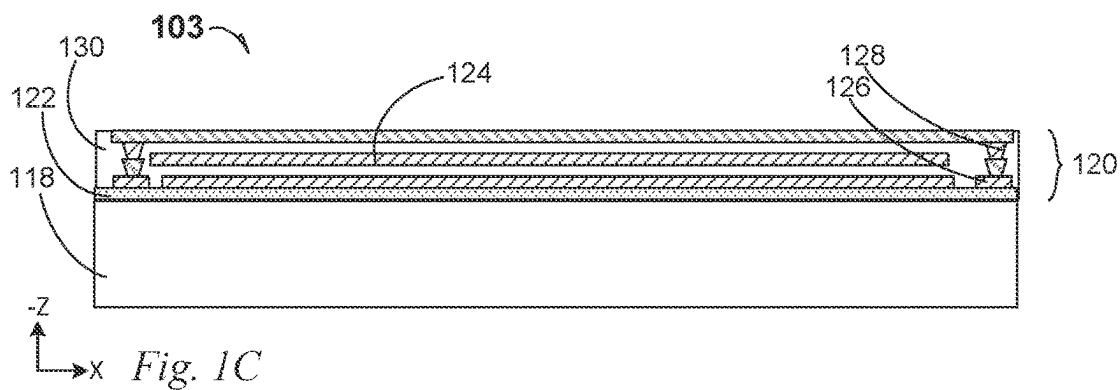
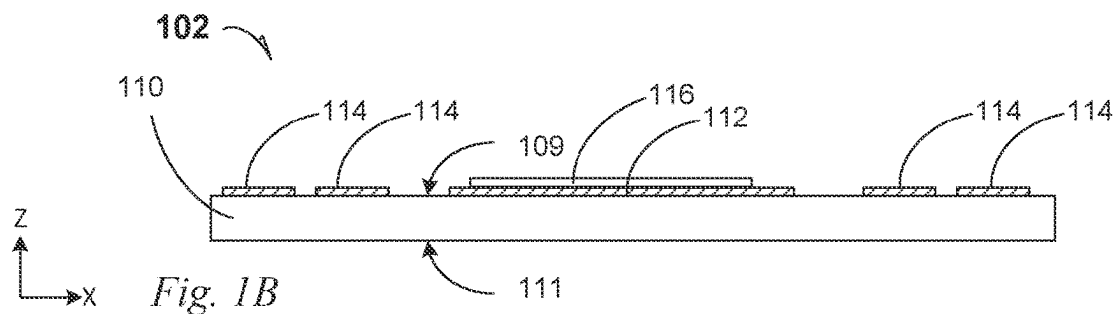
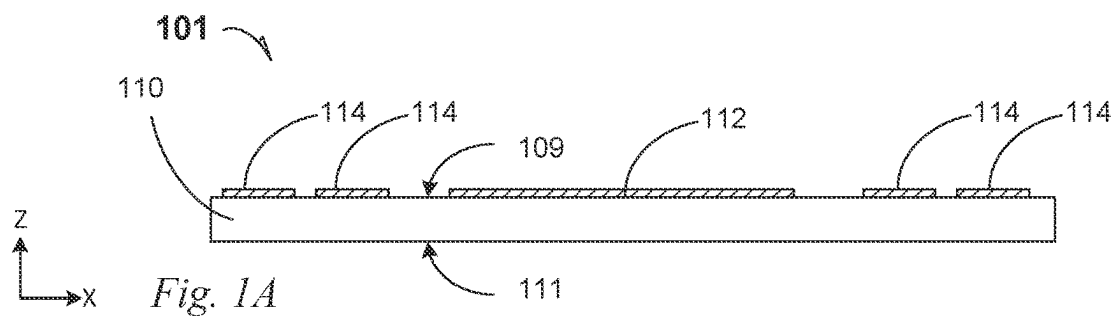
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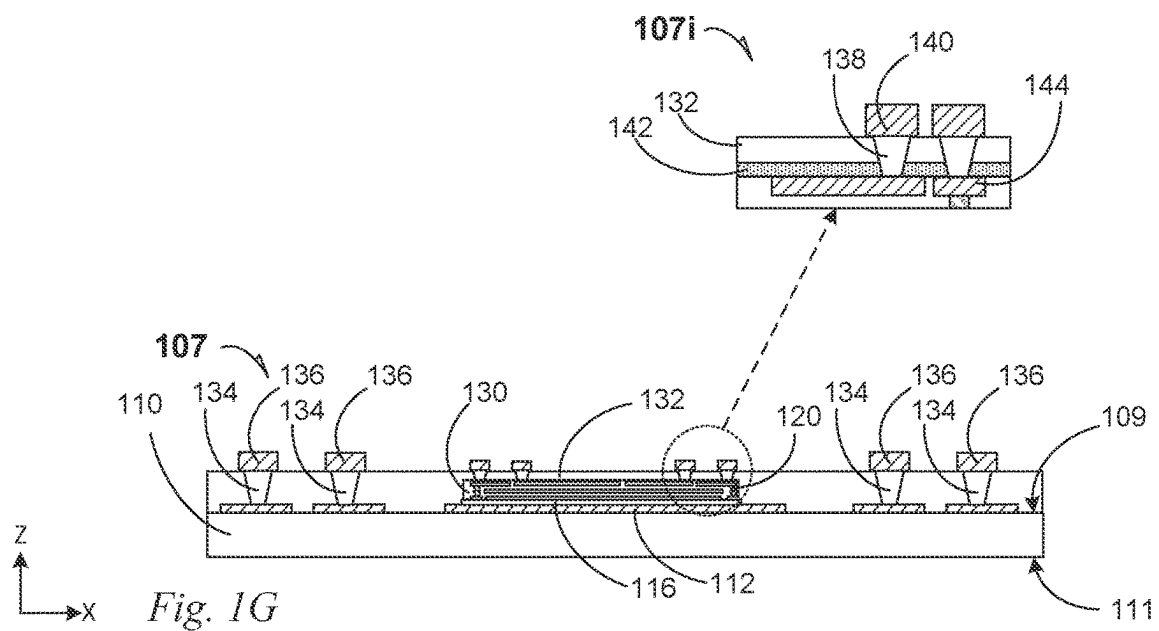
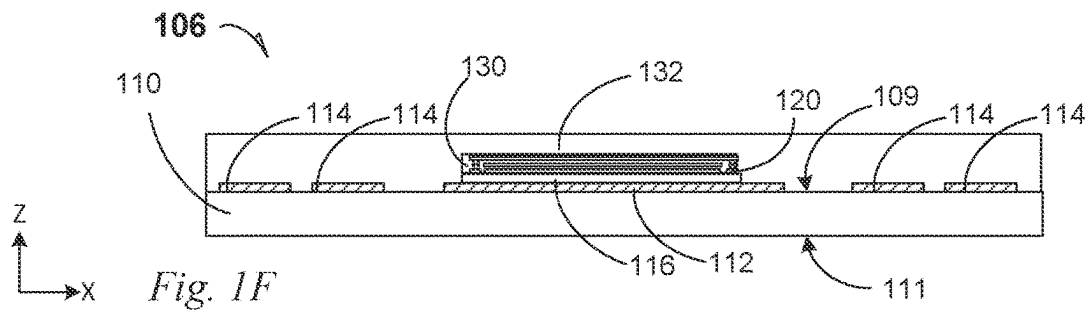
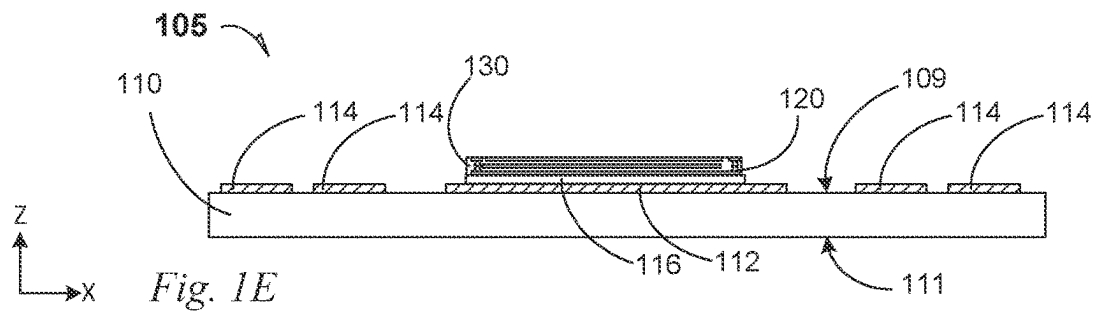
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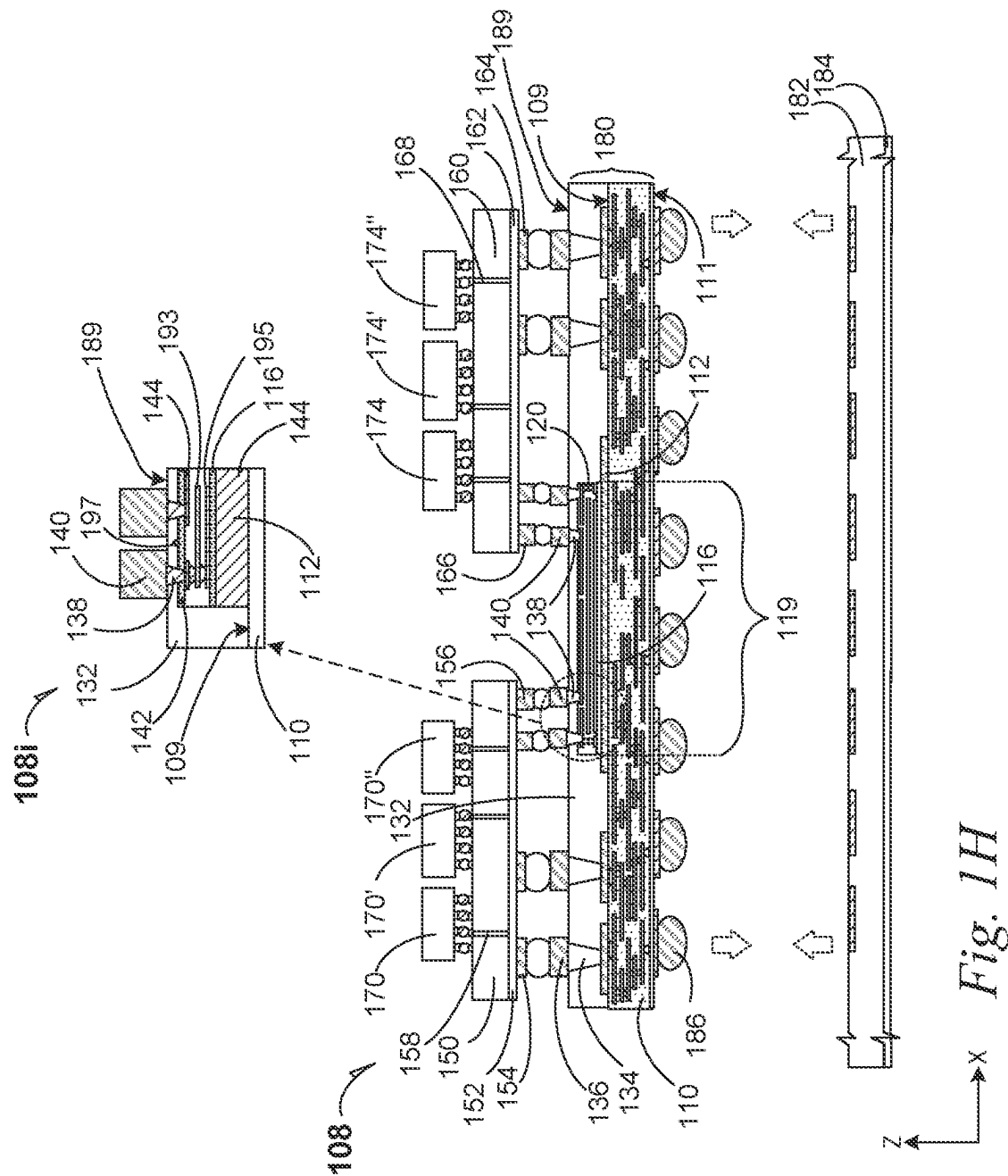
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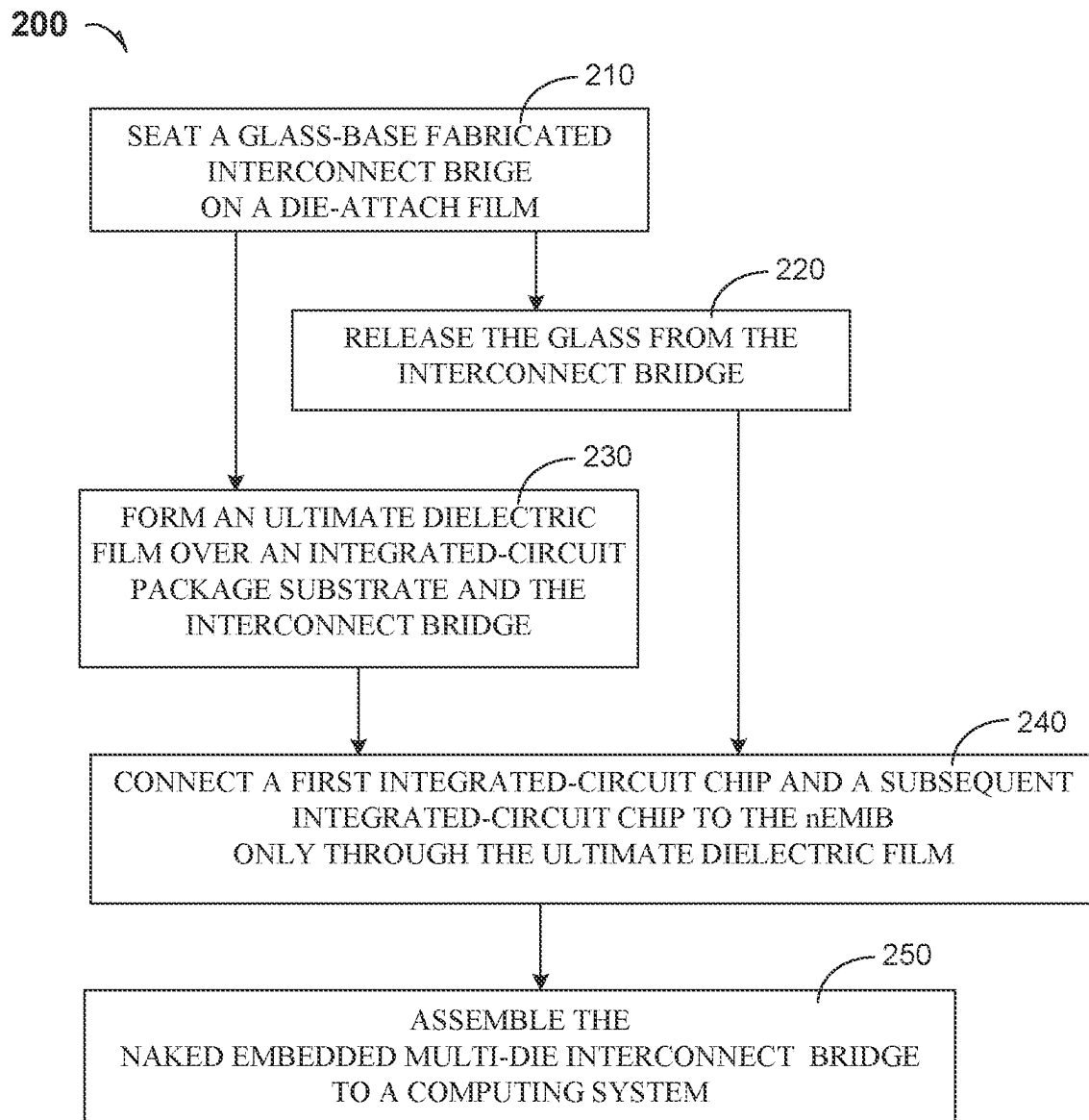
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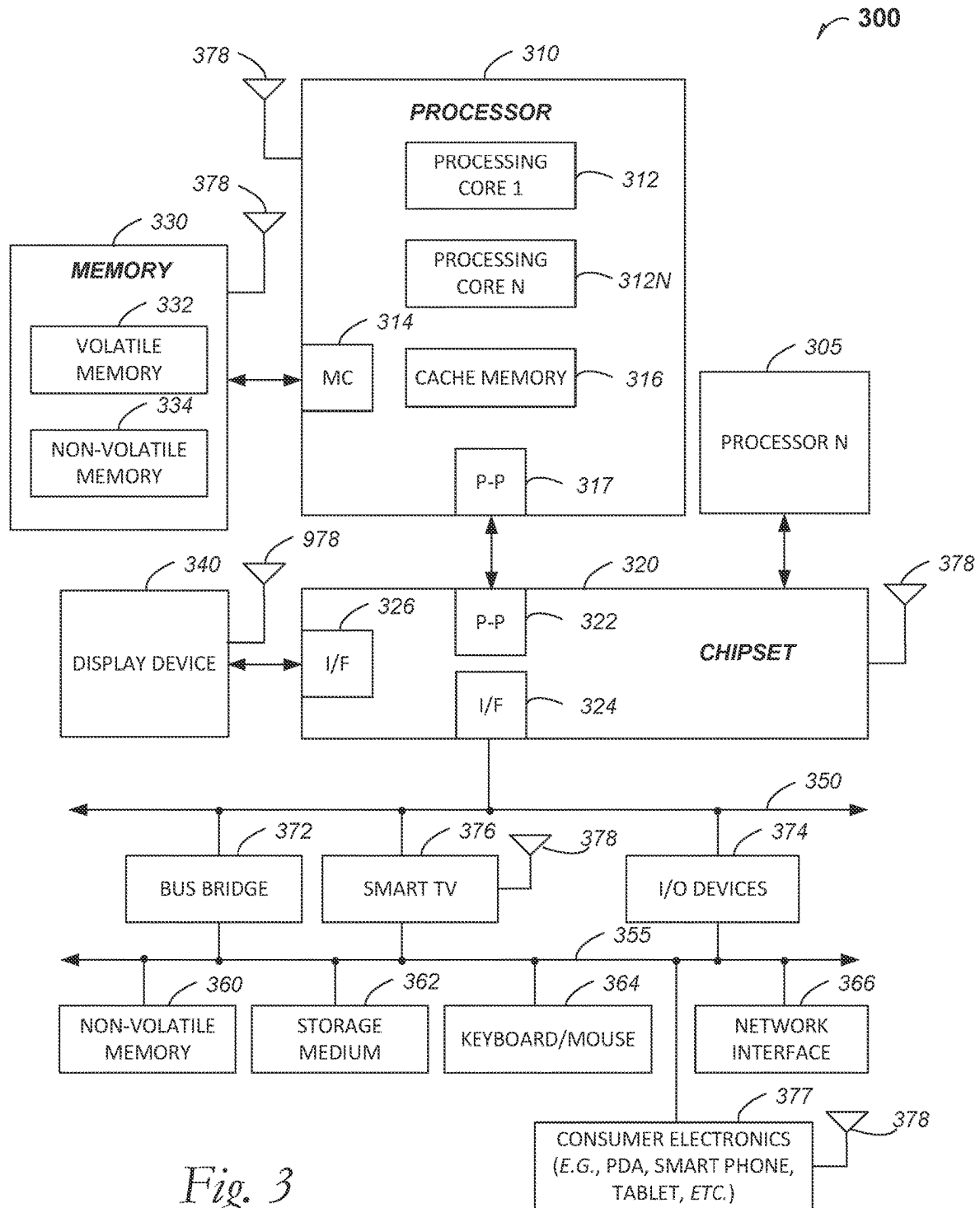
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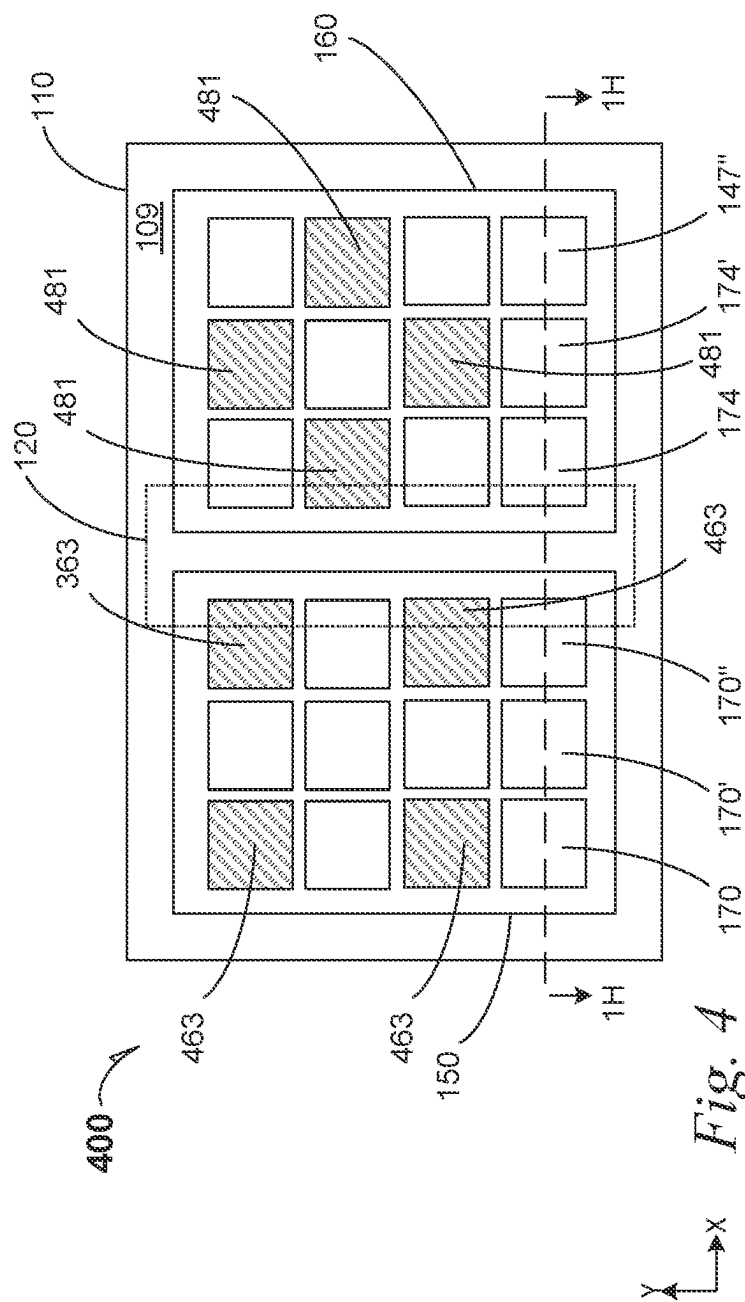






*Fig. 2*







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# STRIPPED REDISTRUBUTION-LAYER FABRICATION FOR PACKAGE-TOP EMBEDDED MULTI-DIE INTERCONNECT BRIDGE

## CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/716,937, filed Apr. 8, 2022, which is a continuation of U.S. patent application Ser. No. 17/200,700, filed Mar. 12, 2021, now U.S. Pat. No. 11,658,111, issued May 23, 2023, which is a continuation of U.S. patent application Ser. No. 16/384,348, filed on Apr. 15, 2019, now U.S. Pat. No. 10,998,262, issued May 4, 2021, the entire contents of which are hereby incorporated by reference herein.

## FIELD

This disclosure relates to embedded multi-chip interconnect bridges that are seated near the die side of integrated-circuit device packages.

## BACKGROUND

Integrated circuit miniaturization during interconnecting, experiences package real estate budget issues.

## BRIEF DESCRIPTION OF THE DRAWINGS

Disclosed embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings where like reference numerals may refer to similar elements, in which:

FIG. 1A is a cross-section of an integrated-circuit package substrate during assembly according to an embodiment;

FIG. 1B is a cross-section elevation of the integrated-circuit package substrate depicted in FIG. 1A after further assembly according to an embodiment;

FIG. 1C is a cross-section elevation of a stripped embedded multi-die interconnect bridge during assembly on a glass substrate according to an embodiment;

FIG. 1D is a cross-section elevation of the integrated-circuit package substrate depicted in FIG. 1B and the glass-mounted stripped embedded multi-die interconnect bridge depicted in FIG. 1C after further processing according to an embodiment;

FIG. 1E is a cross-section elevation of the integrated-circuit package substrate depicted in FIG. 1D and after further processing according to an embodiment;

FIG. 1F is a cross-section elevation of the integrated-circuit package substrate depicted in FIG. 1E after further processing according to an embodiment;

FIG. 1G is a cross-section elevation of the integrated-circuit package substrate depicted in FIG. 1F after further processing according to an embodiment;

Inset 107i illustrates formation of the bridge-via corridors includes penetrating the build-up film, as well as penetrating a bridge polyimide film in order to reach, e.g. a bridge bond pad according to several embodiments;

FIG. 1H is a cross-section elevation of an integrated-circuit device package that is assembled from the integrated-circuit package and the processed build-up film depicted in FIG. 1G and Inset 107i according to several embodiments;

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Inset 108i depicts the stripped embedded-multi-die interconnect bridge as laid out in a three-level-trace and bond-pad configuration according to an embodiment;

FIG. 2 is a process flow diagram according to several embodiments;

FIG. 3 is included to show an example of a higher-level device application for the disclosed embodiments; and

FIG. 4 is a top plan and partial cut-away of the computing system depicted in FIG. 1H according to several embodiments.

## DETAILED DESCRIPTION

A silicon bridge interconnect is seated just below the top solder-resist layer, after fabricating the bridge interconnect on a glass substrate, and removing the glass substrate. Fabrication of the interconnect layers is done in an inverted configuration compared to that of fabricating an existing silicon bridge interconnect. Stripping of the glass substrate, from the interconnect layers allows for a useful low Z-height of the interconnect bridge where only the interconnect materials remain, and embedding the “stripped” interconnect bridge just below the top solder-resist layer, saves valuable interconnect layers in the package substrate, below the interconnect bridge; at least two copper layers. Stripping of the glass substrate from the interconnect layers of the stripped bridge, also allows for thinner upper layers in the dielectric of the package substrate, which improves signal referencing. Consequently, the stripped embedded multi-die interconnect bridge (sEMIB) allows an integrated-circuit package substrate to retain, e.g. a 3-2-3 package-layer count, instead of a larger 4-2-4 package-layer count. The sEMIB may also be referred to as a stripped redistribution layer (sRDL).

An integrated circuit is fabricated in a substrate that may be semiconductive, such as silicon, doped silicon, and III-V material combinations. Other semiconductive materials may be used such as semiconductive carbon in nanotube configurations. After fabrication, the integrated circuit may be singulated from an array of integrated circuits, into an integrated-circuit chip, or IC chip.

FIG. 1A is a cross-section elevation 101 of an integrated-circuit package substrate during assembly according to an embodiment. A build-up film 110 is used as a basis for forming a metallic plate 112 and bond pads 114, among other structures, for connecting at least two integrated circuits (see FIG. 1H) through a stripped embedded multi-die interconnect bridge (sEMIB). In an embodiment, patterning of the metallic plate 112 and the bond pads 114 is done by patterning the structures 112 and 114 from a single copper-containing layer.

In an embodiment, the build-up film 110 is a partially completed integrated-circuit package substrate 110 with a temporary die side 109 that will accept at least two integrated circuits through an sEMIB, and a land side 111 that will be bonded to a board such as a printed wiring board.

FIG. 1B is a cross-section elevation of the integrated-circuit package substrate 101 depicted in FIG. 1A after further assembly according to an embodiment. The integrated-circuit package substrate 102 has been processed by seating a die-attach film 116 on the metallic plate 112, in anticipation of receiving an sEMIB redistribution layer.

FIG. 1C is a cross-section elevation 103 of a stripped embedded multi-die interconnect bridge (sEMIB) 120 during assembly on a glass substrate 118 according to an embodiment. The Z-direction coordinate is inverted com-

pared to orientation of the integrated-circuit package substrate **102** depicted in FIG. 1B.

A glass substrate **118** is used for patterning and forming an EMIB structure **120** by using lithographic techniques and building the EMIB **120** on a release layer **122**. The glass substrate **118** is a semiconductor package-substrate quality structure with a useful flatness and thermal and physical stability for fabrication of silicon interconnect bridges. Techniques for forming silicon EMIBs on semiconductive material, include to fabricate a "silicon bridge," by fabricating the EMIB **120** on the glass substrate **118**, followed by stripping the glass substrate **118**.

In an embodiment, the EMIB **120** includes traces, e.g. **124**, bond pads, e.g. **126**, and vias, e.g. **128**, with an organic matrix **130** that is several build-up dielectric layers. As illustrated and in an embodiment, a three-trace-layer redistribution layer **120** has been fabricated.

FIG. 1D is a cross-section elevation of the integrated-circuit package substrate **102** depicted in FIG. 1B and the glass-mounted EMIB **103** depicted in FIG. 1C after further processing according to an embodiment. The glass substrate **118** and the EMIB **120** are inverted compared to the depiction in FIG. 1C.

In an embodiment, the EMIB **120** is affixed to the die-attach film **116**, and the release layer **122** is being irradiated by ultraviolet light **121**, to allow the glass substrate **118** to be removed, as well as the release layer **122**. Patterning includes transmitting light energy through the inorganic substrate **118**.

FIG. 1E is a cross-section elevation of the integrated-circuit package substrate **104** depicted in FIG. 1D and after further processing according to an embodiment. The integrated-circuit package substrate **105** has been processed by removing the release layer **122** and stripping the glass substrate **118** (see FIG. 1D) in preparation further building up and connecting the substrate **110** to at least two integrated circuits through the stripped redistribution layer **120**.

FIG. 1F is a cross-section elevation of the integrated-circuit package substrate **105** depicted in FIG. 1E after further processing according to an embodiment. The integrated-circuit package substrate **106** has been processed by forming a build-up film **132** in preparation for forming contact vias for both the integrated-circuit package substrate as well as for the stripped embedded multi-die interconnect bridge (sEMIB) **120**. In an embodiment, the build-up film **132** is a single solder-resist material that is curable by useful light wavelengths, whether a positive photoresist or a negative photoresist. Accordingly, The Z-height of the plate **112** and the bond pads **114** is about 12 micrometer ( $\mu\text{m}$ ), the die-attach film is about 5  $\mu\text{m}$ , the sEMIB **120** is about 10  $\mu\text{m}$ , and the portion of the build-up film **132** above the sEMIB **120** is about 5-10  $\mu\text{m}$ .

FIG. 1G is a cross-section elevation of the integrated-circuit package substrate **106** depicted in FIG. 1F after further processing according to an embodiment. The integrated-circuit package substrate **107** has been processed by opening contact corridors in the build-up film **132**, and filling package vias **134** to the bond pads **114** (see FIG. 1F), as well as forming package copper studs **136** that contact the package vias **134**.

Further processing includes opening bridge-via corridors in the build-up film **132**, and filling bridge vias **138** (see the inset **107i**) as well as bridge copper studs **140**.

The inset **107i** illustrates formation of the bridge-via corridors includes penetrating the build-up film **132**, as well as penetrating a bridge polyimide film **142** in order to reach, e.g. a bridge bond pad **144**. The bridge polyimide film **142**

is part of the sEMIB **120** that remains after the EMIB **120** (see FIG. 1D) is separated from the glass substrate **118** and the release layer **122**.

Removing the release layer **122** and the glass substrate **118** (see FIG. 1D) is done in preparation further building up and connecting the integrated-circuit package substrate **110** to at least two integrated circuits through the sEMIB **120**.

FIG. 1H is a cross-section elevation of an integrated-circuit device package **108** that is assembled from the integrated-circuit package substrate **110** and the processed build-up film **132** depicted in FIG. 1G and Inset **107i** according to several embodiments.

The integrated-circuit package substrate **110**, along with the build-up film **132**, form a perimeter of an integrated-circuit package substrate **180** that carries the stripped embedded multi-die interconnect bridge (sEMIB) **120**. Characteristic of an sEMIB embodiment, the sEMIB **120** is attached to the integrated-circuit package substrate **110** with the die-attach film **116**, and essentially only the buildup film **132** covers the sEMIB **120** as it is in the ultimate build-up layer of the integrated-circuit package substrate **180**. The sEMIB **120** leaves useful printed-wiring-board real estate within the integrated-circuit package substrate **110**, where it is available below the sEMIB **120**. Further, a die side **189** of the integrated-circuit package substrate **180**, only covers the sEMIB **120**, which is seated on the temporary die side **109** and attached with the die-attach film **116**. Accordingly, no remaining glass or otherwise, extends into the useful real estate of the integrated-circuit package substrate **110**.

In an embodiment, the sEMIB **120** is configured to connect with a first integrated circuit chip **150** and a subsequent integrated circuit chip **160**. Bridge bond pads **156** and **166** couple to the sEMIB **120**, through the bridge vias **138** (two reference lines) as well as bridge copper studs **140** (two reference lines).

Each of the respective first and subsequent integrated circuit chips **150** and **160**, is also coupled to the integrated-circuit package substrate **180** in substrate vias **134**, that communicate to the land side **111**.

Accordingly, the sEMIB **120** appears as a redistribution layer (RDL) with photolithographically formed traces and vias, with no glass, nor semiconductor substrate remaining, and a die-attach film **116** seating the sEMIB **120** onto a metallic plate **112**, and only the build-up film **132** covering the sEMIB **120** at the die side **189** of the integrated-circuit package substrate **180**.

The copper pillar **136** contacts an electrical bump that contacts a bonding pad **154** that is part of the first integrated-circuit die **150**. Similarly, copper pillar contacts an electrical bump that contacts a bonding pad **164** that is part of the subsequent integrated-circuit die **160**.

As illustrated, more detail of traces and vias is given generally in the cross-section view of the integrated-circuit package substrate **110**, including immediately below the footprint **119** of the sEMIB **120**, where traces and vias do not appear any less densely, nor differently arrayed in the integrated-circuit package substrate **110** within the footprint **119**, than in any other region of the integrated-circuit package substrate **110**. Specific trace and via density is selected depending upon useful design rules and connections between integrated circuit devices, passive devices and connections to a board **182**.

In an embodiment after forming electrical bumps **186** on the land side **111**, the integrated-circuit package substrate **110** is seated on a board **182** such as a printed-wiring-board motherboard **182**. In an embodiment, the board **182** includes an external shell **184** that provides both physical and elec-

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trical insulation for devices within the external shell **184**. In an embodiment, the board **182** holds a chipset (see FIG. 3).

In an embodiment, the integrated-circuit device package **108** is a base structure for a disaggregated-die computing system **108** that includes chiplets, e.g. **170**, **170'** and **170"** coupled to the first integrated-circuit die **150**, one chiplet **170** of which is illustrated coupled to the first integrated-circuit die **150** at active devices and metallization **152** by a through-silicon via (TSV) **158**. The copper pillar **136** contacts an electrical bump that contact a bonding pad **145** that is part of the first integrated-circuit die **150**.

Similarly in an embodiment, the integrated-circuit device package **108** is a base structure for a disaggregated-die computing system **108** that includes chiplets **174**, **174'** and **174"** coupled to the subsequent integrated-circuit die **160**, one chiplet **174"** of which is illustrated coupled to the subsequent integrated-circuit die **174"** at active devices and metallization **162** through a TSV **168**.

In an embodiment, the sEMIB **120** is laid out in a three-level-trace and bond-pad configuration, as depicted in an Inset **108i** according to an embodiment. The integrated-circuit package substrate **110** is depicted in part, and the metallic plate **112** is an Nth metal layer in the integrated-circuit package substrate **110**. The Nth metal layer in an embodiment, is the top conductive layer in a 3-2-3 package-layer count for the integrated-circuit substrate **110**.

The bridge vias **138** as well as bridge copper studs **140** are coupled to a signal layer **144**, which is at the bridge bond-pad layer **144** depicted in Inset **7i**. The Nth metal layer **112** is a voltage source, source (VSS) layer, as well as a middle layer **193**, which is also coupled to VSS. As illustrated, a signal layer **195** abuts the die-attach film **116**, and signal integrity is enhanced by location of the signal layer **195**, across from the die-attach film **116** and opposite the Nth metal layer **112** as a VSS layer **112**.

In an embodiment, processing by use of the release layer **122** illustrated in FIG. 1D, leaves a release layer inclusion **197** that can be detected between the polyimide film **142** and the ultimate layer **132**. Such release-layer inclusions **197** can be incidental to stripping the glass substrate **118**, also seen in FIG. 1D.

In an embodiment, the build-up film **132** is a single solder-resist material that is curable by useful light wavelengths, whether it is a positive photoresist or a negative photoresist. Accordingly with the single photoresist build-up film **132** covering the stripped embedded multi-die interconnect bridge **120**, the Z-height of the plate **112** is about 12 micrometer ( $\mu\text{m}$ ), the die-attach film is about 5  $\mu\text{m}$ , the sEMIB **120** is about 10  $\mu\text{m}$ , and the portion of the build-up film **132** above the sEMIB **120** is about 5-10  $\mu\text{m}$ .

In an embodiment, the first integrated-circuit die **150** and at least one chiplet, e.g. **170'** make a disaggregated-die logic processor. In an embodiment, the subsequent integrated-circuit die **160** and at least one chiplet, e.g. **174'** make a disaggregated-die graphics processor.

In an embodiment, the first integrated-circuit die **150** and all chiplets, e.g. **170**, **170'** and **170"**, as well as the subsequent integrated-circuit die **160** and all chiplets, e.g. **174**, **174'** and **174"** make a disaggregated-die logic processor. In an embodiment, the first integrated-circuit die **150** and all chiplets, e.g. **170**, **170'** and **170"**, as well as the subsequent integrated-circuit die **160** and all chiplets, e.g. **174**, **174'** and **174"** make a disaggregated-die graphics processor.

In an embodiment, the sEMIB **120** is made to translate between two or more design-rule geometries, where the first integrated-circuit die **150** has a larger design-rule geometry than that of the subsequent integrated-circuit die **160**. Con-

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sequently as a stripped RDL **120**, the sEMIB **120** translates between at least two different design-rule geometries.

FIG. 2 is a process flow diagram according to several embodiments.

At **210**, the process includes seating a glass-based manufactured interconnect bridge on a die-attach film at a penultimate layer of an integrated-circuit package substrate. In an embodiment, the inorganic base is semiconductive or undoped silicon that is a sacrificial substrate.

At **220**, the process includes releasing the glass from the interconnect bridge.

At **230**, the process includes forming an ultimate dielectric layer on the integrated-circuit package substrate and the interconnect bridge.

At **240**, the process includes connecting a first integrated-circuit chip, and a subsequent integrated-circuit chip to the sEMIB, only through the ultimate dielectric layer.

At **250**, the process includes assembling the sEMIB to a computing system.

FIG. 3 is included to show an example of a higher-level device application for the disclosed embodiments. The stripped embedded multi-die interconnect bridge embodiments may be found in several parts of a computing system. In an embodiment, the stripped embedded multi-die interconnect bridge is part of a communications apparatus such as is affixed to a cellular communications tower. In an embodiment, a computing system **300** includes, but is not limited to, a desktop computer. In an embodiment, a system **300** includes, but is not limited to a laptop computer. In an embodiment, a system **300** includes, but is not limited to a netbook. In an embodiment, a system **300** includes, but is not limited to a tablet. In an embodiment, a system **300** includes, but is not limited to a notebook computer. In an embodiment, a system **300** includes, but is not limited to a personal digital assistant (PDA). In an embodiment, a system **300** includes, but is not limited to a server. In an embodiment, a system **300** includes, but is not limited to a workstation. In an embodiment, a system **300** includes, but is not limited to a cellular telephone. In an embodiment, a system **300** includes, but is not limited to a mobile computing device. In an embodiment, a system **300** includes, but is not limited to a smart phone. In an embodiment, a system **300** includes, but is not limited to an internet appliance. Other types of computing devices may be configured with the microelectronic device that includes stripped embedded multi-die interconnect bridge embodiments.

In an embodiment, the processor **310** has one or more processing cores **312** and **312N**, where **312N** represents the Nth processor core inside processor **310** where N is a positive integer. In an embodiment, the electronic device system **300** using a stripped embedded multi-die interconnect bridge embodiment that includes multiple processors including **310** and **305**, where the processor **305** has logic similar or identical to the logic of the processor **310**. In an embodiment, the processing core **312** includes, but is not limited to, pre-fetch logic to fetch instructions, decode logic to decode the instructions, execution logic to execute instructions and the like. In an embodiment, the processor **310** has a cache memory **316** to cache at least one of instructions and data for the stripped embedded multi-die interconnect bridge in the system **300**. The cache memory **316** may be organized into a hierarchal structure including one or more levels of cache memory.

In an embodiment, the processor **310** includes a memory controller **314**, which is operable to perform functions that enable the processor **310** to access and communicate with memory **330** that includes at least one of a volatile memory

**332** and a non-volatile memory **334**. In an embodiment, the processor **310** is coupled with memory **330** and chipset **320**. In an embodiment, the chipset **320** is part of a system-in-package with a stripped embedded multi-die interconnect bridge depicted in FIGS. 1H, and Inset **108i**. The processor **310** may also be coupled to a wireless antenna **378** to communicate with any device configured to at least one of transmit and receive wireless signals. In an embodiment, the wireless antenna interface **378** operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

In an embodiment, the volatile memory **332** includes, but is not limited to, Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS Dynamic Random Access Memory (RDRAM), and/or any other type of random access memory device. The non-volatile memory **334** includes, but is not limited to, flash memory, phase change memory (PCM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), or any other type of non-volatile memory device.

The memory **330** stores information and instructions to be executed by the processor **310**. In an embodiment, the memory **330** may also store temporary variables or other intermediate information while the processor **310** is executing instructions. In the illustrated embodiment, the chipset **320** connects with processor **310** via Point-to-Point (PtP or P-P) interfaces **317** and **322**. Either of these PtP embodiments may be achieved using a stripped embedded multi-die interconnect bridge embodiment as set forth in this disclosure. The chipset **320** enables the processor **310** to connect to other elements in a stripped embedded multi-die interconnect bridge embodiment in a system **300**. In an embodiment, interfaces **317** and **322** operate in accordance with a PtP communication protocol such as the Intel® QuickPath Interconnect (QPI) or the like. In other embodiments, a different interconnect may be used.

In an embodiment, the chipset **320** is operable to communicate with the processor **310**, **305N**, the display device **340**, and other devices **372**, **376**, **374**, **360**, **362**, **364**, **366**, **377**, etc. The chipset **320** may also be coupled to a wireless antenna **378** to communicate with any device configured to at least do one of transmit and receive wireless signals.

The chipset **320** connects to the display device **340** via the interface **326**. The display **340** may be, for example, a liquid crystal display (LCD), a plasma display, cathode ray tube (CRT) display, or any other form of visual display device. In an embodiment, the processor **310** and the chipset **320** are merged into a stripped embedded multi-die interconnect bridge in a computing system. Additionally, the chipset **320** connects to one or more buses **350** and **355** that interconnect various elements **374**, **360**, **362**, **364**, and **366**. Buses **350** and **355** may be interconnected together via a bus bridge **372** such as at least one stripped embedded multi-die interconnect bridge package apparatus embodiment. In an embodiment, the chipset **320**, via interface **324**, couples with a non-volatile memory **360**, a mass storage device(s) **362**, a keyboard/mouse **364**, a network interface **366**, smart TV **376**, and the consumer electronics **377**, etc.

In an embodiment, the mass storage device **362** includes, but is not limited to, a solid state drive, a hard disk drive, a universal serial bus flash memory drive, or any other form of computer data storage medium. In one embodiment, the network interface **366** is implemented by any type of well-known network interface standard including, but not limited

to, an Ethernet interface, a universal serial bus (USB) interface, a Peripheral Component Interconnect (PCI) Express interface, a wireless interface and/or any other suitable type of interface. In one embodiment, the wireless interface operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

While the modules shown in FIG. 3 are depicted as separate blocks within the embedded magnetic inductor and a stripped embedded multi-die interconnect bridge package in a computing system **300**, the functions performed by some of these blocks may be integrated within a single semiconductor circuit or may be implemented using two or more separate integrated circuits. For example, although cache memory **316** is depicted as a separate block within processor **310**, cache memory **316** (or selected aspects of **316**) can be incorporated into the processor core **312**.

In an embodiment, a single processor **310** includes stripped embedded multi-die interconnect bridge embodiments with chiplets, such as one, all or more than the chiplets **170** through **170''** and optionally one, all or more than the chiplets **174** through **174''**, making up the single processor **310**.

Where useful, the computing system **300** may have a broadcasting structure interface such as for affixing the apparatus to a cellular tower.

FIG. 4 is a top plan **400** and partial cut-away of the computing system **108** depicted in FIG. 1H according to several embodiments. The disaggregated-die computing system **108** depicted in FIG. 1H is seen at the cross-section line 1H-1H. The first integrated-circuit die **150** is seen in below a chiplet array, as well as the subsequent integrated-circuit die **160**. The first-die chiplets **170**, **170'** and **170''** are seen above the first die **150** according to an embodiment. Similarly, the subsequent-die chiplets **174**, **174'** and **174''** are seen above the subsequent die **160** according to an embodiment.

In an embodiment, a 3x4 array of chiplet spaces is configured on the backside surface of the first integrated-circuit chip **150**, but four of the spaces are taken up by heat slugs **463** to facilitate heat removal from the first chip **150** and into a heat sink such as an integrated heat spreader that contacts the heat slugs. In an embodiment, all first chiplet spaces are taken up by integrated-circuit chiplets. Similarly in an embodiment, a 3x4 array of chiplet spaces is configured on the backside surface of the subsequent chip **160**, but four of the spaces are taken up by heat slugs **481** to facilitate heat removal from the subsequent die **160** and into the same heat sink that contacts the heat slugs **463** according to an embodiment. In an embodiment, all subsequent chiplet spaces are taken up by integrated-circuit chiplets.

As illustrated, different useful patterns for heat slugs **463** and **481** are applied above the respective first and subsequent integrated-circuit chips **150** and **160**, depending upon heat-extraction usefulness.

To illustrate the stripped embedded multi-die interconnect bridge embodiments and methods disclosed herein, a non-limiting list of examples is provided herein:

Example 1 is an integrated-circuit device package, comprising: a first integrated-circuit die on an integrated-circuit package substrate, wherein the integrated-circuit package substrate includes a die side and a land side; an interconnect bridge on a penultimate layer of the integrated-circuit package substrate; a single dielectric layer over the interconnect bridge, wherein the single dielectric layer includes the die side as an ultimate layer, is penetrated by a bridge via that contacts the interconnect bridge, and wherein the single

dielectric layer is penetrated by a package via that is electrically coupled to the land side.

In Example 2, the subject matter of Example 1 optionally includes wherein the interconnect bridge is on a die-attach film that is on the penultimate layer of the integrated-circuit package substrate.

In Example 3, the subject matter of any one or more of Examples 1-2 optionally include wherein the interconnect bridge is on a die-attach film that is on the penultimate layer of the integrated-circuit package substrate, and wherein the interconnect bridge has three conductive layers with a bottom conductive layer contacting the die-attach film, and a top conductive layer being contacted by the bridge via.

In Example 4, the subject matter of any one or more of Examples 1-3 optionally include micrometer.

In Example 5, the subject matter of any one or more of Examples 1-4 optionally include a first integrated-circuit die that is coupled to the bridge via that contacts the interconnect bridge, and that is coupled to the package via.

In Example 6, the subject matter of any one or more of Examples 1-5 optionally include a first integrated-circuit die that is coupled to the bridge via, wherein the bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the first integrated-circuit die; and wherein the first integrated-circuit die is coupled to the land side through the package via, that contacts a copper pillar that contacts an electrical bump.

In Example 7, the subject matter of any one or more of Examples 1-6 optionally include a first integrated-circuit die that is coupled to the bridge via, wherein the bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the first integrated-circuit die; wherein the first integrated-circuit die is coupled to the land side through the package via, that contacts a copper pillar that contacts an electrical bump; a subsequent integrated-circuit die that is coupled to subsequent bridge via, wherein the subsequent bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the subsequent integrated-circuit die; and wherein the subsequent integrated-circuit die is coupled to the land side through a subsequent package via, that contacts a copper pillar that contacts an electrical bump.

In Example 8, the subject matter of any one or more of Examples 1-7 optionally include wherein the single dielectric layer covering is also over and contacting the penultimate layer.

In Example 9, the subject matter of Example 8 optionally includes wherein the interconnect bridge is on a die-attach film that is on the penultimate layer of the integrated-circuit package substrate.

In Example 10, the subject matter of any one or more of Examples 8-9 optionally include wherein the interconnect bridge is on a die-attach film that is on the penultimate layer of the integrated-circuit package substrate, and wherein the interconnect bridge has three conductive layers with a bottom conductive layer contacting the die-attach film.

In Example 11, the subject matter of any one or more of Examples 1-10 optionally include a first integrated-circuit die that is coupled to the bridge via that contacts the interconnect bridge, and that is coupled to the package via.

In Example 12, the subject matter of any one or more of Examples 1-11 optionally include a first integrated-circuit die that is coupled to the bridge via, wherein the bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the first integrated-circuit die; and wherein the first integrated-circuit die is coupled to the land

side through the package via, that contacts a copper pillar that contacts an electrical bump.

In Example 13, the subject matter of any one or more of Examples 1-12 optionally include wherein the ultimate layer, contacts a release-layer inclusion, the interconnect bridge further including a polyimide film that is penetrated by the bridge via, and wherein the release-layer inclusion is between the polyimide film and the ultimate layer.

Example 14 is a computing system, comprising: an integrated-circuit device package including a die side and a land side; a first integrated circuit on the integrated-circuit package substrate die side; an interconnect bridge on a penultimate layer of the integrated-circuit package substrate; a single dielectric layer covering over the interconnect bridge, wherein the single dielectric layer includes the die side as an ultimate layer, is penetrated by a bridge via that contacts the interconnect bridge, and wherein the single dielectric layer is penetrated by a package via that is electrically coupled to the land side; a first integrated-circuit die that is coupled to the bridge via, wherein the bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the first integrated-circuit die; wherein the first integrated-circuit die is coupled to the land side through the package via, that contacts a copper pillar that contacts an electrical bump; a subsequent integrated-circuit die that is coupled to subsequent bridge via, wherein the subsequent bridge via contacts a copper bridge pad, that contacts an electrical bump, that contacts the subsequent integrated-circuit die; wherein the subsequent integrated-circuit die is coupled to the land side through a subsequent package via, that contacts a copper pillar that contacts an electrical bump; and wherein the first and subsequent integrated-circuit dice are part of a chipset.

In Example 15, the subject matter of Example 14 optionally includes more than one chiplet on the first integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the first integrated-circuit die through a through-silicon via.

In Example 16, the subject matter of any one or more of Examples 14-15 optionally include more than one chiplet on the first integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the first integrated-circuit die through a through-silicon via; and more than one chiplet on the subsequent integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the subsequent integrated-circuit die through a through-silicon via.

In Example 17, the subject matter of any one or more of Examples 14-16 optionally include more than one chiplet on the first integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the first integrated-circuit die through a through-silicon via; more than one chiplet on the subsequent integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the subsequent integrated-circuit die through a through-silicon via; and wherein at least one of the more than one chiplet on the first integrated-circuit, at least one of the more than one chiplet on the subsequent integrated-circuit die, and at least one of the first and subsequent integrated-circuit dice comprise a disaggregated logic processor.

In Example 18, the subject matter of any one or more of Examples 14-17 optionally include more than one chiplet on the first integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the first integrated-circuit die through a through-silicon via; more than one chiplet on the subsequent integrated-circuit die at a backside, wherein one of the more than one chiplet com-

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municates to the subsequent integrated-circuit die through a through-silicon via; and wherein at least one of the more than one chiplet on the first integrated-circuit, wherein at least one of the more than one chiplet on the subsequent integrated-circuit die, and at least one of the first and subsequent integrated-circuit dice comprise a disaggregated graphics processor.

In Example 19, the subject matter of any one or more of Examples 14-18 optionally include more than one chiplet on the first integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the first integrated-circuit die through a through-silicon via; more than one chiplet on the subsequent integrated-circuit die at a backside, wherein one of the more than one chiplet communicates to the subsequent integrated-circuit die through a through-silicon via; wherein at least one of the more than one chiplet on the first integrated-circuit, wherein at least one of the more than one chiplet on the subsequent integrated-circuit die, and at least one of the first and subsequent integrated-circuit dice comprise a disaggregated logic processor, and wherein at least one of the more than one chiplet on the first integrated-circuit, wherein at least one of the more than one chiplet on the subsequent integrated-circuit die, and at least one of the first and subsequent integrated-circuit dice comprise a disaggregated graphics processor.

Example 20 is a process of forming an interconnect bridge, comprising: patterning a metallization on an inorganic substrate, including patterning three metallization layers including vias and traces; stripping the inorganic substrate after seating the metallization on a die-attach film of an integrated-circuit package substrate; and forming a single dielectric layer over the metallization and the integrated-circuit package substrate.

In Example 21, the subject matter of Example 20 optionally includes wherein patterning includes patterning with a top layer first and a bottom layer last, followed by inverting the metallization when seating on the die-attach film.

In Example 22, the subject matter of any one or more of Examples 20-21 optionally include connecting a first and a subsequent integrated-circuit die to the metallization through the single dielectric layer.

In Example 23, the subject matter of any one or more of Examples 20-22, optionally include wherein patterning the metallization on the inorganic substrate includes transmitting light energy through the inorganic substrate.

The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

In the event of inconsistent usages between this document and any documents so incorporated by reference, the usage in this document controls.

In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is

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used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

Method examples described herein can be machine or computer-implemented at least in part. Some examples can include a computer-readable medium or machine-readable medium encoded with instructions operable to configure an electrical device to perform methods as described in the above examples. An implementation of such methods can include code, such as microcode, assembly language code, a higher-level language code, or the like. Such code can include computer readable instructions for performing various methods. The code may form portions of computer program products. Further, in an example, the code can be tangibly stored on one or more volatile, non-transitory, or non-volatile tangible computer-readable media, such as during execution or at other times. Examples of these tangible computer-readable media can include, but are not limited to, hard disks, removable magnetic disks, removable optical disks (e.g., compact disks and digital video disks), magnetic cassettes, memory cards or sticks, random access memories (RAMs), read only memories (ROMs), and the like.

The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description as examples or embodiments, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the disclosed embodiments should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

The invention claimed is:

1. An integrated-circuit device package, comprising: an interconnect bridge in a dielectric layer, the interconnect bridge having a top side and a bottom side opposite the top side, a first side between the top side and the bottom side, and a second side between the top side and the bottom side, wherein the dielectric layer is in contact with the top side and the first side of the interconnect bridge, the interconnect bridge comprising first bridge vias and second bridge vias, wherein an uppermost surface of the first bridge vias and the

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- second bridge vias is at a same level as an uppermost surface of the dielectric layer;
- a first via and a second via in and in contact with the dielectric layer, the first via and the second via laterally spaced apart from the first side of the interconnect bridge in a cross-sectional perspective, and the first via and the second via having an uppermost surface at a same level as the uppermost surface of the dielectric layer;
  - a first copper stud coupled to the first via, and a second copper stud coupled to the second via;
  - a third via and a fourth via in and in contact with the dielectric layer, the third via and the fourth via laterally spaced apart from the second side of the interconnect bridge in the cross-sectional perspective, and the third via and the fourth via having an uppermost surface at a same level as the uppermost surface of the dielectric layer;
  - a third copper stud coupled to the third via, and a fourth copper stud coupled to the fourth via;
  - a first integrated circuit chip coupled to the first bridge vias of the interconnect bridge, the first integrated circuit chip electrically coupled to the first copper stud and the second copper stud;
  - a second integrated circuit chip having a top side and a bottom side, the bottom side of the second integrated circuit chip coupled to the second bridge vias of the interconnect bridge, the second integrated circuit chip electrically coupled to the third copper stud and the fourth copper stud, and the second integrated circuit chip having through-silicon vias;
  - a third integrated circuit chip above the top side of the second integrated circuit chip, the third integrated circuit chip coupled to the through-silicon vias of the second integrated circuit chip; and
  - a package substrate beneath the interconnect bridge, the first via, the second via, the third via and the fourth via, the package substrate electrically coupled to the first via, the second via, the third via and the fourth via.
2. The interconnect bridge of claim 1, wherein the first integrated circuit chip is coupled to the first bridge vias of the interconnect bridge by a plurality of first bridge copper studs, and wherein the second integrated circuit chip is coupled to the second bridge vias of the interconnect bridge by a plurality of second bridge copper studs.
3. The integrated-circuit device package of claim 1, further comprising:
- a fourth integrated circuit chip coupled to the top side of the second integrated circuit chip.
4. The integrated-circuit device package of claim 3, further comprising:
- a fifth integrated circuit chip coupled to the top side of the second integrated circuit chip.
5. The integrated-circuit device package of claim 4, wherein the fourth integrated circuit chip is laterally adjacent to the third integrated circuit chip, and the fifth integrated circuit chip is laterally adjacent to the fourth integrated circuit chip.
6. The integrated-circuit device package of claim 4, wherein each of the third integrated circuit chip, the fourth integrated circuit chip, and the fifth integrated circuit chip is directly attached to the top side of the second integrated circuit chip.
7. The integrated-circuit device package of claim 4, further comprising:
- a sixth integrated circuit chip coupled to a top side of the first integrated circuit chip.

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8. The integrated-circuit device package of claim 1, wherein the interconnect bridge is in direct contact with the dielectric layer, and wherein each of the first, second, third and fourth vias is in direct contact with the dielectric layer.
9. The integrated-circuit device package of claim 1, further comprising:
- a die attach film coupled to the bottom side of the interconnect bridge.
10. The integrated-circuit device package of claim 9, further comprising:
- a heat slug coupled to the die attach film.
11. An integrated-circuit device package, comprising:
- an interconnect bridge in a dielectric layer, wherein the dielectric layer is in contact with a top side and the first side of the interconnect bridge, the interconnect bridge comprising a plurality of bridge vias, wherein an uppermost surface of the plurality of bridge vias is at a same level as an uppermost surface of the dielectric layer;
  - a first plurality of vias in and in contact with the dielectric layer, the first plurality of vias laterally spaced apart from a first side of the interconnect bridge, and the first plurality of vias having an uppermost surface at a same level as the uppermost surface of the dielectric layer;
  - a first plurality of conductive structures coupled to the first plurality of vias;
  - a second plurality of vias in and in contact with the dielectric layer, the second plurality of vias laterally spaced apart from a second side of the interconnect bridge, and the second plurality of vias having an uppermost surface at a same level as the uppermost surface of the dielectric layer;
  - a second plurality of conductive structures coupled to the second plurality of vias;
  - a first chip coupled to a first portion of the plurality of bridge vias, the first chip electrically coupled to the first plurality of conductive structures;
  - a second chip coupled to a second portion of the plurality of bridge vias, the second chip electrically coupled to the second plurality of conductive structures, and the second chip having through vias;
  - a third chip above the top side of the second chip, the third chip coupled to the through vias of the second chip; and
  - a package substrate beneath the interconnect bridge, the first plurality of vias, and the second plurality of vias.
12. The integrated-circuit device package of claim 11, further comprising:
- a fourth chip coupled to the top side of the second chip.
13. The integrated-circuit device package of claim 12, further comprising:
- a fifth chip coupled to the top side of the second chip.
14. The integrated-circuit device package of claim 13, wherein the fourth chip is laterally adjacent to the third chip, and the fifth chip is laterally adjacent to the fourth chip.
15. The integrated-circuit device package of claim 13, further comprising:
- a sixth chip coupled to a top side of the first chip.
16. An integrated-circuit device package, comprising:
- an interconnect bridge in a dielectric layer, the interconnect bridge having a top side and a bottom side opposite the top side, a first side between the top side and the bottom side, and a second side between the top side and the bottom side, wherein the dielectric layer is in contact with the top side and the first side of the interconnect bridge, the interconnect bridge comprising a plurality of bridge vias, wherein an uppermost surface of the plurality of bridge vias is at a same level as an uppermost surface of the dielectric layer;

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vias in and in contact with the dielectric layer, the vias laterally spaced apart from the first side of the interconnect bridge and laterally spaced apart from the second side of the interconnect bridge in a cross-sectional perspective, and the vias having an uppermost surface at a same level as the uppermost surface of the dielectric layer;

copper studs coupled to the vias;

a first integrated circuit chip coupled to a first portion of the plurality of bridge vias and to a first portion of the copper studs;

a second integrated circuit chip having a top side and a bottom side, the bottom side of the second integrated circuit chip coupled to a second portion of the plurality of bridge vias and to a second portion of the copper studs, and the second integrated circuit chip having through-silicon vias;

a third integrated circuit chip above the top side of the second integrated circuit chip, the third integrated circuit chip coupled to the through-silicon vias of the second integrated circuit chip; and

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a package substrate beneath the interconnect bridge and the vias, the package substrate electrically coupled to the vias.

**17.** The interconnect bridge of claim **16**, wherein the first integrated circuit chip is coupled to the first portion of the plurality of bridge vias by a first portion of bridge copper studs, and wherein the second integrated circuit chip is coupled to the second portion of the plurality of bridge vias by a second portion of bridge copper studs.

**18.** The integrated-circuit device package of claim **16**, further comprising:

a fourth integrated circuit chip coupled to the top side of the second integrated circuit chip.

**19.** The integrated-circuit device package of claim **18**, further comprising:

a fifth integrated circuit chip coupled to the top side of the second integrated circuit chip.

**20.** The integrated-circuit device package of claim **19**, wherein the fourth integrated circuit chip is laterally adjacent to the third integrated circuit chip, and the fifth integrated circuit chip is laterally adjacent to the fourth integrated circuit chip.

\* \* \* \* \*